acting on the vehicle. If the air pressure is high or low from its specified pressure then there will be negative effects which will be caused by the tyre to the vehicle. Hence, based on the ply rating, vehicle speed and the load of the vehicle, the air pressure is mentioned or recommended by the vehicle manufacturer.

## 5.14 Selection of the tyre

The tyre selection for any vehicle is based on the following considerations

1. It should work for a long time.
2. The price of the tyre should be low and should have good quality.
3. The tyre must have the required tread pattern and ply rating for the specific application.
4. The tyre must have the good quality rubber mixture.

## 5.15 Tyre wear

The tyre wear occurs due to the following reasons,

1. Due to the pressure of the air filled in the tyre is lower or higher than its specified pressure
2. Due to the reasons of suddenly and frequently applying the brake.
3. If the wheel is not fixed properly on the axle.
4. If there are wear on the bearings of the wheels.
5. If the vehicle carry's more loads.
6. The wheel alignment angles such as caster and camber are not set correctly as per the recommendations.
7. If the load on the vehicle is not balanced.

## 5.16 Maintenance of Tyre

The long life of the tyre could be possible by proper maintenance of the tyres. The following are the points to be followed for good tyre maintenance.

1. The required amount of air should be filled and maintained correctly.
2. The air pressure has to be checked by using the pressure gauge at least once in a week.
3. If there is any grease or oil present in tyre, it must be cleaned.
4. The tyre should not be overloaded.
5. Often braking the vehicle or unnecessarily braking during high speeds of the vehicle have to be avoided.
6. The vehicle should not be turned quickly when running on bends.
7. The tyre must be checked for its position on the axle whether it is placed on the rim with straight and stable condition or not.

## 5.17 Tyre repairs and rectification methods

| Repairs | Repairing methods |
|---|---|
| Tyre wear on one side | Correct the wheel alignment |
| Tyre wear at the center | Maintain the tyre pressure. |
| The tyre pressure reduced often | Valve in the tube should be corrected |
| Explosion or scratches occurring in tyre. | Should carry specified load |
| Tyre wear occurring soon | Correct the wheel alignment |



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